

Database Management Systems

CPS 3740

Lecture 04

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Outline

- Chapter 6
 - Aggregation functions
 - Group by
- Do a lot of exercises on command line
- Lab 1 and project
 - Start working on them ASAP

Chapter 6

**SQL – Structure Query Language
Data Manipulation**

SELECT Statement - Aggregates

- ISO standard defines five aggregate functions:

COUNT returns number of values in specified column.

SUM returns sum of values in specified column.

AVG returns average of values in specified column.

MIN returns smallest value in specified column.

MAX returns largest value in specified column.

- No space between function name and ()
- Examples: MAX(salary), AVG(salary), COUNT(SID)

SELECT Statement - Aggregates

- Each operates on a single column of a table and returns a single value.
- COUNT, MIN, and MAX apply to numeric and non-numeric fields, but SUM and AVG may be used on numeric fields only.
- Apart from COUNT(*), each function eliminates nulls first and operates only on remaining non-null values.
- “*” can be only used in COUNT() function

SELECT Statement - Aggregates

- **COUNT(*)** counts **all rows** of a table, regardless of whether nulls or duplicate values occur.
 - The result is different if count() on a column that has NULL values.
- Can use **DISTINCT** before column name to eliminate duplicates.
- **DISTINCT** has no effect with MIN/MAX, but may have with SUM/AVG.

SELECT Statement - Aggregates

- Aggregate functions can be used only in **SELECT list** and **in HAVING** clause.
- If SELECT list includes an aggregate function and there is no GROUP BY clause, SELECT list cannot reference a column out with an aggregate function. For example, the following is illegal:

SELECT staffNo, COUNT(salary) FROM dreamhome.Staff;

Example 6.13 Use of COUNT(*)

- How many properties cost more than 350 per month to rent?

SELECT COUNT(*) AS myCount
FROM dreamhome.PropertyForRent
WHERE rent > 350;

myCount
5

propertyNo	street	city	postcode	type	rooms	rent	ownerNo	staffNo	branchNo
PA14	16 Holh...	Aberdeen	AB7 5SU	House	6	650.00	CO46	SA9	B007
PG16	5 Novar ...	Glasgow	G12 9AX	Flat	4	450.00	CO93	SG14	B003
PG21	18 Dale ...	Glasgow	G12	House	5	600.00	CO87	SG37	B003
PG36	2 Manor...	Glasgow	G32 4QX	Flat	3	375.00	CO93	SG37	B003
PG4	6 Lawre...	Glasgow	G11 9QX	Flat	3	350.00	CO40	NULL	B003
PL94	6 Argyll ...	London	NW2	Flat	4	400.00	CO87	SL41	B005

Class exercises – aggregation with *

- Use **dreamhome.Test**
- Compare the results for the following statements
 - `SELECT count(value) from dreamhome.Test;`
 - `SELECT count(id) from dreamhome.Test;`
 - `SELECT count(*) from dreamhome.Test;`
- The following statements are **illegal (syntax error)**:
 - `SELECT sum(*) from dreamhome.Test;`
 - `SELECT max(*) from dreamhome.Test;`
 - `SELECT avg(*) from dreamhome.Test;`

Example 6.14 Use of COUNT(DISTINCT)

- How many different properties viewed between '2013-04-20' and '2013-05-31'?

SELECT COUNT(DISTINCT propertyNo) AS myCount
FROM dreamhome.Viewing
WHERE viewDate **BETWEEN**
'2013-04-20' AND '2013-05-31';

myCount
3

clientNo	propertyNo	viewDate	comment
CR56	PA14	2013-05-24	too small
CR56	PG36	2013-04-28	
CR56	PG4	2013-05-26	
CR62	PA14	2013-04-14	no dining room
CR76	PG4	2013-04-20	too remote

Class exercises – aggregation, distinct

- Use **dreamhome.Staff**
- Show the maximum salary
 - > `SELECT max(salary) from dreamhome.Staff;`
 - > `SELECT max(distinct salary) from dreamhome.Staff;`
 - > `SELECT distinct max(salary) from dreamhome.Staff;`
 - These 3 statements should have the same results.
- Show the number of salary
 - > `SELECT count(salary) from dreamhome.Staff;`
 - > `SELECT count(distinct salary) from dreamhome.Staff;`
 - > `SELECT distinct count(salary) from dreamhome.Staff;`
 - These 3 statements may have different results.

Example 6.15 Use of COUNT and SUM

- Find number of Managers and sum of their salaries.

```
SELECT COUNT(staffNo) AS myCount,  
        SUM(salary) AS mySum  
FROM dreamhome.Staff  
WHERE position = 'Manager';
```

myCount	mySum
2	54000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Example 6.16 Use of MIN, MAX, AVG

- Find minimum, maximum, and average staff salary.

```
SELECT MIN(salary) AS myMin,  
       MAX(salary) AS myMax,  
       AVG(salary) AS myAvg  
FROM dreamhome.Staff;
```

myMin	myMax	myAvg
9000.00	30000.00	17000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Class exercises – aggregation, NULL

- Use **dreamhome.Test**
- Compare the results for the following statements
 1. `SELECT max(value) from dreamhome.Test;`
 2. `SELECT count(value) from dreamhome.Test;`
 3. `SELECT count(distinct value) from dreamhome.Test;`
 4. `SELECT count(id) from dreamhome.Test;`
 5. `SELECT count(distinct id) from dreamhome.Test;`
 6. `SELECT sum (value) from dreamhome.Test;`
 7. `SELECT sum(distinct value) from dreamhome.Test;`
 8. `SELECT count(value), max(value), sum(value) from dreamhome.Test where value is NULL;`

SELECT Statement - Grouping

- Use **GROUP BY** clause to get sub-totals.
- **SELECT** and **GROUP BY** closely integrated: each item in SELECT list must be *single-valued per group*, and SELECT clause may only contain:
 - column names
 - aggregate functions
 - constants
 - expression involving combinations of the above.
- **Can GROUP BY more than one field.**

SELECT Statement - Grouping

- All column names in SELECT list must appear in GROUP BY clause unless name is used only in an aggregate function.
- If WHERE is used with GROUP BY, WHERE is applied first, then groups are formed from remaining rows satisfying predicate.

Example 6.17 Use of GROUP BY

- Find number of staff in each branch and their total salaries.

```
SELECT branchNo,  
        COUNT(staffNo) AS myCount,  
        SUM(salary) AS mySum  
FROM dreamhome.Staff  
GROUP BY branchNo  
ORDER BY branchNo;
```

branchNo	myCount	mySum
B003	3	54000.00
B005	2	39000.00
B007	1	9000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Example 6.17 Use of GROUP BY

- Find number of staff in each branch and their total salaries. -- group by two data fields.

```
SELECT branchNo, staffNo,  
        COUNT(staffNo) AS myCount,  
        SUM(salary) AS mySum  
FROM dreamhome.Staff  
GROUP BY branchNo, staffNo  
ORDER BY branchNo;
```

branchNo	staffNo	myCount	mySum
B003	SG14	1	18000.00
B003	SG37	1	12000.00
B003	SG5	1	24000.00
B005	SL21	1	30000.00
B005	SL41	1	9000.00
B007	SA9	1	9000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Class exercises – GROUP BY

- Use **dreamhome.Staff**
- Show the number of staff for each position
- Show the number of staff for each gender in each branch
- Show the number of female staff in each branch

Restricted Groupings – HAVING clause

- **HAVING** clause is designed for **use with GROUP BY** to restrict groups that appear in final result table.
- Similar to **WHERE**, but **WHERE** filters individual rows whereas **HAVING** filters groups.
- **. Column names in HAVING clause must also appear in the GROUP BY list or be contained within an aggregate function**

Example 6.18 Use of HAVING

- For each branch with more than 1 member of staff, find number of staff in each branch and sum of their salaries.

```
SELECT branchNo, COUNT(staffNo) AS myCount,  
        SUM(salary) AS mySum  
FROM dreamhome.Staff  
GROUP BY branchNo  
HAVING COUNT(staffNo) > 1  
ORDER BY branchNo;
```

branchNo	myCount	mySum
B003	3	54000.00
B005	2	39000.00

staffNo	fName	lName	position	sex	DOB	salary	branchNo
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003
SL21	John	White	Manager	M	1945-10-01	30000.00	B005
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005

Self exercise – GROUP BY, HAVING

- Use **dreamhome.Staff**
- Show the number of staff for each position that has at least 2 people
- Show the number of female staff in each branch that has 2 female staff
 - The output should be 2 if a branch that has have 2 female and 1 male staff
 - The branch should not be in the output if it has 2 male and 1 female

Self exercise – GROUP BY, HAVING

Answers and discussions

- Show the number of staff for each position that has at least 2 people

Select position, count(staffno) from dreamhome.Staff group by position having count(staffno)>=2;

- Or use alias

Select position, count(staffno) as ct from dreamhome.Staff group by position having ct >=2;

- Show the number of female staff in each branch that has at least 2 female staff

Select branchno, count(staffno) from dreamhome.Staff where sex='F' group by branchno having count(staffno) >= 2;

- The following is illegal (syntax error)

Select branchno, count(staffno) from dreamhome.Staff group by branchno having count(staffno) >= 2 where sex='F';

Multi-Table Queries

- Can use subqueries provided result columns come from same table.
- If result columns come from more than one table must use a join.
- To perform join, include more than one table in FROM clause.
- Use **comma** as separator and typically include WHERE clause to specify join column(s).

Multi-Table Queries Alias

- Also possible to use an alias for a table named in FROM clause.
- Alias is separated from table name with a space.
- Alias can be used to qualify column names when there is ambiguity.

Example 6.24 Simple Join

- List names of all clients who have viewed a property along with any comment supplied.

SELECT c.clientNo, fName, lName, propertyNo, comment
FROM Client c, Viewing v **WHERE** c.clientNo = v.clientNo;

clientNo	propertyNo	viewDate	comment
CR56	PA14	2013-05-24	too small
CR56	PG36	2013-04-28	
CR56	PG4	2013-05-26	
CR62	PA14	2013-04-14	no dining room
CR76	PG4	2013-04-20	too remote

clientNo	fName	lName	propertyNo	comment
CR56	Aline	Stewart	PA14	too small
CR56	Aline	Stewart	PG36	
CR56	Aline	Stewart	PG4	
CR62	Mary	Tregear	PA14	no dining room
CR76	John	Kay	PG4	too remote

clientNo	fName	lName	telNo	preType	maxRent	eMail
client	fName	lName	telNo	preType	0.00	email
CR56	Aline	Stewart	0141-8...	Flat	350.00	astewar...
CR62	Mary	Tregear	01224-...	Flat	600.00	maryt@...
CR74	Mike	Richie	01475-...	House	750.00	mritchie...
CR76	John	Kay	0207-7...	Flat	425.00	john.kay...

Example 6.24 Simple Join

- Only those rows from both tables that have identical values in the clientNo columns (**c.clientNo = v.clientNo**) are included in result.
- Equivalent to equi-join in relational algebra.

clientNo	fName	lName	propertyNo	comment
CR56	Aline	Stewart	PG36	too small
CR56	Aline	Stewart	PA14	
CR56	Aline	Stewart	PG4	
CR62	Mary	Tregear	PA14	no dining room
CR76	John	Kay	PG4	too remote

Exercise – JOIN 2 tables

- Use Staff and Branch tables in the dreamhome
- Show the staff name and their working city
 - Staff name is in Staff table
 - City is in Branch table
 - Need to join Staff and Branch tables with a joining condition - the same branchno

Select fname, lname, city from Staff, Branch ;

→ meaningless

Select fname, lname, city from Staff, Branch where
Staff.branchno = Branch.branchno;

- Use alias for tables to reduce length

Select fname, lname, city from Staff **s**, Branch **b** where
s.branchno = **b**.branchno;

Exercise – JOIN with condition

- Use Staff and Branch tables in the dreamhome
- Show the **female** staff name and their working city

Select fname, lname, city from Staff **s**, Branch **b** where
s.branchno = b.branchno and s.sex='F' ;

- Show the staff name and their working city in London or Glasgow

Select fname, lname, city from Staff **s**, Branch **b** where
**s.branchno = b.branchno and (b.city='London' or
b.city='Glasgow');**

- Note: Must have (), and the “or” should be inside ().

Alternative JOIN Constructs

- SQL provides alternative ways to specify joins:

FROM Client c JOIN Viewing v ON c.clientNo = v.clientNo

FROM Client JOIN Viewing USING clientNo

FROM Client NATURAL JOIN Viewing

- In each case, FROM replaces original FROM and WHERE. However, first produces table with two identical clientNo columns.
- Be careful when you mix JOIN ON and WHERE

Example – JOIN ON

- The SQL query
 - Select fname, lname, city from Staff s, Branch b **where** s.branchno = b.branchno and s.sex='F' ;
- is the same as
 - Select fname, lname, city from Staff s **join** Branch b **on** s.branchno = b.branchno and s.sex='F' ;
 - Select fname, lname, city from Staff s **join** Branch b **on** s.branchno = b.branchno **where** s.sex='F' ;
 - Select fname, lname, city from Staff s **join** Branch b **where** s.branchno = b.branchno and s.sex='F' ;
- But the following query is **illegal** (syntax error)
 - Select fname, lname, city from Staff s, Branch b **on** s.branchno = b.branchno and s.sex='F' ;

Example 6.25 Sorting a join

- For each branch, list the staff numbers and names of staff who manage properties, and properties they manage.

```
SELECT s.branchNo, s.staffNo, fName, lName,  
       propertyNo  
FROM Staff s, PropertyForRent p  
WHERE s.staffNo = p.staffNo  
ORDER BY s.branchNo, s.staffNo, propertyNo;
```

branchNo	staffNo	fName	lName	propertyNo
B003	SG14	David	Ford	PG16
B003	SG37	Ann	Beech	PG21
B003	SG37	Ann	Beech	PG36
B005	SL41	Julie	Lee	PL94
B007	SA9	Mary	Howe	PA14

SQL self-join (the table joins itself)

- Question – how to find all staff fname and direct supervisor's fname.

```
MariaDB [dreamhome]> select * from Staff1;
```

staffNo	fName	lName	position	sex	DOB	salary	branchNo	superno
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007	SG5
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003	SG5
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003	SG5
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003	SL21
SL21	John	White	Manager	M	1945-10-01	30000.00	B005	NULL
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005	SL21

staff	supervisor
Mary	Susan
David	Susan
Ann	Susan
Susan	John
Julie	John

Output

SELF JOIN – in two SQL queries

- Use dreamhome.Staff
- Find the supervisor's first name for "Susan".
 - Without using self join, we need to find Susan's superno.
 - **Select superno from Staff1 where fname='Susan';**
- And use the superno as the staffno to find the first name
 - **select fname from Staff1 where staffno='SL21';**

staffNo	fName	lName	position	sex	DOB	salary	branchNo	superno
ABC1	New	Enginer	Intern	M	2000-01-01	9000.00	B002	SL21
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007	SG5
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003	SG5
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003	SG5
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003	SL21
SL21	John	White	Manager	M	1945-10-01	30000.00	B005	NULL
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005	SL21
XY1	New	Tester	Assistant	M	1990-01-01	9000.00	B002	SL21

SELF JOIN: 1 – in ONE SQL query

- A table joins itself
 - Give different alias for each table
- Must give meaningful column names
 - Both tables have identical Column names.
- Find the supervisor's first name for "Susan".
 - `select m.fname from Staff1 e, Staff1 m where e.superno=m.staffno and e.fname='Susan';`

staffNo	fName	lName	position	sex	DOB	salary	branchNo	superno
ABC1	New	Enginer	Intern	M	2000-01-01	9000.00	B002	SL21
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007	SG5
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003	SG5
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003	SG5
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003	SL21
SL21	John	White	Manager	M	1945-10-01	30000.00	B005	NULL
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005	SL21
XY1	New	Tester	Assistant	M	1990-01-01	9000.00	B002	SL21

SELF JOIN: 2 – in ONE SQL query

- Find the supervisor's first name for every Staff.

- select e.fname as employee,
m.fname as manager
from Staff1 e, Staff1 m
where e.superno=m.staffno;

(Q) John is not in the employee column,
how to include John in the employee?

employee	manager
New	John
Mary	Susan
David	Susan
Ann	Susan
Susan	John
Julie	John
New	John

staffNo	fName	lName	position	sex	DOB	salary	branchNo	superno
ABC1	New	Enginer	Intern	M	2000-01-01	9000.00	B002	SL21
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007	SG5
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003	SG5
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003	SG5
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003	SL21
SL21	John	White	Manager	M	1945-10-01	30000.00	B005	NULL
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005	SL21
XY1	New	Tester	Assistant	M	1990-01-01	9000.00	B002	SL21

SELF JOIN: 3 – in ONE SQL query

- If we also want to include John in the employee (**left join**, will discuss it in later lectures)
 - select e.fname as employee,
m.fname as mananger
from Staff1 e **left join** Staff1 m
on e.superno=m.staffno;

employee	mananger
New	John
Mary	Susan
David	Susan
Ann	Susan
Susan	John
John	NULL
Julie	John
New	John

staffNo	fName	lName	position	sex	DOB	salary	branchNo	superno
ABC1	New	Enginer	Intern	M	2000-01-01	9000.00	B002	SL21
SA9	Mary	Howe	Assistant	F	1970-02-19	9000.00	B007	SG5
SG14	David	Ford	Supervisor	M	1958-03-24	18000.00	B003	SG5
SG37	Ann	Beech	Assistant	F	1960-11-10	12000.00	B003	SG5
SG5	Susan	Brand	Manager	F	1940-06-03	24000.00	B003	SL21
SL21	John	White	Manager	M	1945-10-01	30000.00	B005	NULL
SL41	Julie	Lee	Assistant	F	1965-06-13	9000.00	B005	SL21
XY1	New	Tester	Assistant	M	1990-01-01	9000.00	B002	SL21

Class exercises – aggregation function, table join

Step1: Write your queries for the following questions.

- Given the Staff table (staffNo, fName, lName, position, sex, DOB, salary, branchNo), and Branch table (branchNo, street, city, postcode), write and explain an SQL query to find
 1. The number of staff members and the maximum salary in each branchNo.
 2. The number of female staff members in each city.
 3. The 'branchNo' where no staff members, using the 'NOT IN' clause.
 4. The 'branchNo' with more than 2 female staff members in the city of Glasgow.

ChatGPT SQL exercises

Step2: Ask ChatGpt: Copy/paste the above statements into ChatGPT.

Step3: Discussion:

- Who has the correct answers?
- Execute the queries in the database, and verify the results.
- Can you explain your own queries?

Note: ChatGPT is for assistance only. It is not allowed for use during exams. Therefore, you should learn how to solve the problems on your own.

More in class SQL exercises

- Refer to the pdf file
- Work on questions 17 – 43
 - SELECT with aggregations and group by

Home Exercise

- Unix commands: ls, mkdir, cd, chmod, mv, ps,
- SQL commands: desc, show, use
- The following SQL & the examples in textbook/slides
 - SELECT
 - AS
 - GROUP BY, HAVING
 - ORDER BY
 - **Aggregates functions:** MAX, COUNT, MIN, AVG, SUM

Leetcode.com SQL exercise

- 181. Employees Earning More Than Their Managers
 - <https://leetcode.com/problems/employees-earning-more-than-their-managers/>
- Given the Employee table, write a SQL query that finds out employees who earn more than their managers. For the above table, Joe is the only employee who earns more than his manager.

Id	Name	Salary	ManagerId
1	Joe	70000	3
2	Henry	80000	4
3	Sam	60000	NULL
4	Max	90000	NULL

Employee
Joe

Output

Leetcode.com SQL exercise

- 182. Duplicate Emails
 - <https://leetcode.com/problems/duplicate-emails/submissions/>
- Write a SQL query to find all duplicate emails in a table named Person.

Id	Email
1	a@b.com
2	c@d.com
3	a@b.com

Email
a@b.com

Output

Reading Assignment

- 6.3 DML – NULL, NOT NULL
- 6.3 DML – Aggregate functions
- 6.3 DML – ORDER BY, DESC, ASC
- 6.3 DML – GROUP BY, HAVING
- Exercises 6.10 – 6.18
- 29.5 HTTP Cookie
- PHP SQL injection

Conclusion

- Exercise the SQL query on database sever
 - Try the examples in textbook/slides
 - HAVING
 - GROUP BY
- Aggregates functions: MAX, COUNT, MIN, AVG, SUM
- Work on Lab1 and project